

WASHINGTON COUNTY AP, PA, USA

WMO: 725117

Lat: 40.133N

Lon: 80.283W

Elev: 1185

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	7.9	-7.7	3.8	5.1	-2.0	5.2	11.5	24.2	35.6	20.8	30.6	4.5	240	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	88.3	71.8	85.6	71.0	83.2	69.6	74.8	83.7	73.2	81.7	71.7	80.4	6.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	124.3	80.6	70.4	116.8	78.1	69.6	113.7	77.1	39.0	83.8	37.5	81.6	36.2	80.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.0	16.8	14.6	DB	-5.7	91.1	6.7	3.6	-10.5	93.6	-14.4	95.7	-18.2	97.7	-23.1	100.3	(4)
			WB	-6.5	77.5	7.0	1.9	-11.5	78.8	-15.6	79.9	-19.5	81.0	-24.6	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	28.2	31.2	38.7	50.2	59.4	67.3	70.9	69.5	63.0	52.6	41.4	33.5
	DBStd	17.49	13.03	11.75	10.78	9.39	8.19	6.20	5.06	4.94	6.95	8.96	9.45	10.49
	HDD50	2611	686	534	380	111	15	0	0	0	2	73	288	522
	HDD65	5848	1142	946	815	450	214	50	11	16	118	400	708	978
	CDD50	2827	9	8	31	118	305	519	648	604	392	154	30	9
	CDD65	586	0	0	1	6	39	118	194	155	58	15	0	0
	CDH74	5393	0	1	14	113	448	1101	1661	1316	577	159	3	0
	CDH80	1425	0	0	1	15	78	301	492	354	141	43	0	0
Wind	WSAvg	5.4	7.3	6.8	6.7	6.7	5.3	4.5	3.9	3.4	3.7	4.7	5.9	6.1
Precipitation	PrecAvg	40.2	3.3	2.5	3.3	3.3	4.2	4.1	4.1	3.5	3.4	2.6	2.8	3.0
	PrecMax	53.5	6.8	5.0	5.9	5.1	6.7	7.1	7.3	6.7	8.9	5.3	5.2	6.9
	PrecMin	32.4	1.7	1.1	1.5	1.5	1.7	1.5	2.0	1.9	1.4	0.8	0.7	1.7
	PrecStd	5.4	1.4	0.9	1.3	0.9	1.3	1.5	1.5	1.3	1.7	1.1	1.1	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.2	66.2	75.1	82.0	84.5	89.9	91.1	90.7	88.1	85.7	72.2	63.9
		MCWB	55.3	54.9	58.6	62.2	68.2	70.8	74.7	71.9	70.5	69.4	59.0	55.8
	2%	DB	58.8	61.0	69.8	77.4	81.9	86.2	88.2	87.6	83.8	78.6	66.3	60.6
		MCWB	52.8	52.0	55.0	59.6	65.9	70.8	72.6	72.1	69.7	65.4	55.5	54.2
	5%	DB	54.0	54.9	63.3	72.9	79.4	83.6	84.4	83.9	81.1	72.9	63.2	55.1
		MCWB	49.0	47.4	51.6	57.5	65.2	70.0	71.1	70.4	67.6	61.4	52.8	49.3
	10%	DB	46.4	48.2	56.9	68.2	75.2	81.2	82.1	81.5	77.0	68.5	57.5	51.5
		MCWB	41.9	42.8	48.2	55.2	63.6	68.4	69.7	68.6	65.3	60.1	50.1	46.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.3	58.3	60.2	65.5	72.2	75.2	77.9	75.8	74.6	71.6	61.5	58.8
		MCDB	63.0	64.0	72.3	78.2	81.4	84.7	88.4	85.8	82.6	82.4	67.6	62.8
	2%	WB	53.7	53.9	56.9	62.0	69.4	73.3	75.4	74.0	71.6	67.5	57.4	55.1
		MCDB	57.9	58.6	66.3	72.9	78.7	82.5	84.1	82.7	80.0	74.1	63.8	58.7
	5%	WB	48.9	48.4	53.4	59.4	67.4	71.8	73.3	72.3	69.5	64.0	54.3	50.9
		MCDB	53.1	55.1	60.3	68.9	76.1	81.2	81.2	80.7	76.9	71.1	61.1	54.3
	10%	WB	42.3	43.2	49.5	57.0	65.2	70.0	71.6	70.8	67.5	60.4	50.9	45.8
		MCDB	47.0	49.1	57.6	66.5	72.9	78.5	79.9	78.8	74.5	67.3	57.3	49.7
Mean Daily Temperature Range	5% DB	MDBR	16.4	18.2	20.4	23.7	22.8	21.6	21.1	21.6	23.0	21.8	19.1	16.0
		MCDBR	20.3	26.3	29.4	30.4	27.2	25.5	23.9	24.8	26.8	28.2	26.1	22.2
		MCWBR	16.1	19.5	18.3	16.7	14.9	12.9	11.7	12.3	14.1	16.4	17.8	17.7
	5% WB	MCDBR	19.0	23.8	24.8	25.0	22.1	21.3	20.3	20.1	21.7	23.4	22.6	19.9
		MCWBR	15.9	19.3	17.4	15.4	12.6	11.6	10.8	11.0	12.0	14.1	17.2	17.7
Clear-Sky Solar Irradiance	taub	0.302	0.323	0.348	0.380	0.425	0.456	0.467	0.469	0.395	0.363	0.320	0.300	
	taud	2.464	2.387	2.392	2.332	2.212	2.163	2.150	2.124	2.315	2.405	2.504	2.511	
	Ebn at Noon	275	283	287	284	272	262	258	253	267	265	266	267	
	Edh at Noon	25	31	34	39	45	47	48	47	37	30	24	22	
All-Sky Solar Radiation	RadAvg	491	760	1073	1447	1677	1828	1815	1659	1361	920	626	421	
	RadStd	50	113	97	137	140	144	119	96	139	96	82	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+112

Nomenclature: See separate page